

LABORATORY RECORD

Course: Full Stack Web Development

Course Code: 23CM4121

Experiment No.: 2

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Date: 30-08-2026

1. TITLE

Webpage Styling with External CSS3, Linear Gradients, Navigation Bar & Flexbox

2. AIM

To structure and style a multi-card webpage using external CSS3 stylesheets, linear gradients, responsive navigation bars, flexbox card grid layout, box shadows, and button hover transitions.

3. OBJECTIVE

The objectives of this experiment are:

- To link external CSS3 stylesheets to HTML5 documents.
 - To apply multi-color linear gradients and text shadows on headers.
 - To style a dark navigation bar with interactive hover color changes.
 - To implement responsive flexbox card layout with borders and shadows.
 - To create button hover color transitions.
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4. THEORY

External CSS separates document structure from visual presentation. CSS3 provides linear gradients, Flexbox for flexible horizontal/vertical alignment (display: flex, justify-content, gap), box shadows for modern UI depth, and pseudo-classes (:hover) for interactive transitions.

Application Flow:

HTML5 Structure -> External CSS Linking -> Gradient Header & Nav -> Flexbox Container -> Card Styling & Hover Effects

5. ALGORITHM / PROCEDURE

1. Create index.html and link external style.css using <link rel='stylesheet'>.
 2. Style header with linear-gradient(to right, #0066cc, #8000ff) and text shadow.
 3. Style navigation bar with dark background and yellow link hover color.
 4. Implement .container with display: flex, justify-content: center, and gap: 20px.
 5. Style .card with white background, rounded border, padding, and box shadow.
 6. Style buttons with green background and red hover color transition.
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7. Style footer with dark background.

8. Verify styling and responsive layout across different screen sizes.

6. PROGRAM

index.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>My Styled Webpage</title>
  <link rel="stylesheet" href="style.css">
</head>
<body>

<header>
  <h1>My Styled Webpage</h1>
</header>

<nav>
  <a href="#">Home</a>
  <a href="#">About</a>
  <a href="#">Services</a>
  <a href="#">Contact</a>
</nav>

<section class="container">

  <div class="card">
    <h2>HTML5</h2>
    <p>HTML5 is used to create the
structure of web pages.</p>
    <button>Read More</button>
  </div>

  <div class="card">
    <h2>CSS3</h2>
    <p>CSS3 is used to style web pages and improve appearance.</p>
    <button>Read More</button>
  </div>

  <div class="card">
    <h2>Web Design</h2>
    <p>HTML and CSS together create
beautiful web applications.</p>
    <button>Read More</button>
  </div>

</section>

<footer>
  <p>© 2026 My Styled Webpage</p>
</footer>

</body>
</html>
```

style.css

```
body{
  margin:0;
  font-family:Arial,sans-serif;
  background:#f2f2f2; }

header{
  background:linear-gradient(to
right,#0066cc,#8000ff);
  color:white;
  text-align:center;
  padding:20px;
  text-shadow:2px 2px
5px gray; }

nav{
  background:#333;
  text-align:center;
  padding:15px; }

nav a{
  color:white;
  text-decoration:none;
```

```
margin:20px;      font-
size:18px;  }

nav a:hover{
color:yellow;  }

.container{
display:flex;      justify-
content:center;
gap:20px;      margin:30px;
}

.card{      background:white;
width:250px;      padding:20px;
border:2px solid gray;
border-radius:15px;      text-
align:center;      box-shadow:5px
5px 10px gray;  }

button{
background:green;
color:white;
border:none;
padding:10px 15px;
border-radius:5px;
cursor:pointer;  }

button:hover{
background:red;  }

footer{
background:black;
color:white;      text-
align:center;
padding:15px;
margin-top:20px;
}
```

7. CODE EXPLANATION

Code / Syntax	Explanation
<link rel='stylesheet' href='style.css'>	Links the external CSS stylesheet to the HTML document.
background: linear-gradient(to right, #0066cc, #8000ff)	Creates a two-color horizontal gradient background on the header.
nav a:hover { color: yellow; }	Changes link font color to yellow on mouse hover.
display: flex; justify-content: center; gap: 20px;	Flexbox container centering child cards with 20px uniform gap.
border-radius: 15px; box-shadow: 5px 5px 10px gray;	Rounds card corners and adds realistic drop shadows.
button:hover { background: red; }	Transitions button color from green to red upon mouse hover.

8. TEST CASES AND EXECUTION DATA

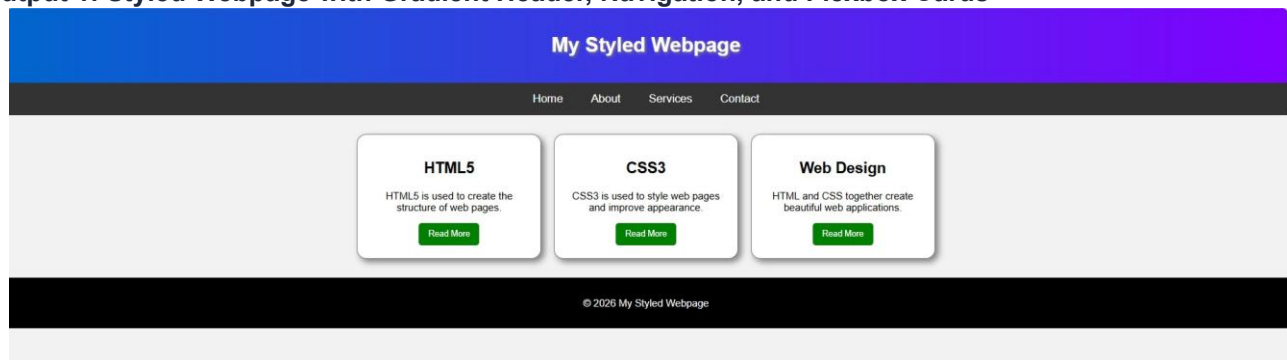
The experiment was executed with the following test cases to verify functionality:

Test Case	Input / Action	Expected Result	Actual Result	Status
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TC-01	CSS Linkage	Open styled page	Header gradient and typography applied correctly	Pass
TC-02	Nav Hover State	Hover over nav links	Link color turns yellow on hover	Pass
TC-03	Flexbox Cards Alignment	Inspect cards container	Three cards aligned in a centered row with 20px gap	Pass
TC-04	Card Box Shadow	Inspect cards	Cards render 15px rounded corners and box shadows	Pass
TC-05	Button Hover Effect	Hover over 'Read More'	Button color transitions from green to red	Pass

9. OUTPUT

Output 1: Styled Webpage with Gradient Header, Navigation, and Flexbox Cards



10. RESULT

Thus, external CSS3 styling with linear gradients, navigation bar hover effects, and flexbox card layout was successfully implemented and verified.